Math 302 Project Overview and Instructions

The aim of the Math 302 project is to apply the topics studied in the Partial Differential Equations& Complex Analysis course to solve various models that employ both Differential Equations and Complex Analysis. These models arise in various fields such as aerodynamics, elasticity, electrodynamics, fluid dynamics, heat transfer, plasma physics, electromagnetics, quantum mechanics, etc.

Groups of **4-5 students** should propose and present a related idea as their final course project. Each group selects a topic to be addressed as a project. The topic idea should be communicated and approved by your instructor.

Project Milestones

Week	8th	10th	12th	14th
Task	• Form a team • Point/title • Short abstract	• Proposal • Analytical/ Numerical	• Simulation • Draft report	• Presentation • Final report
% grade	10%	10%	10%	70%

Notes

- **Form a team, select a topic and write an abstract by week 8. (10 points)**
- **Solve the proposed problem using analytical and/or numerical methods and writing a proposal that contains abstract, introduction, methodology, and references about the employed technique by week 10. (10 points)**
- **Use ready-made SW packages such as Mathematica and/or Matlab for problem solving/simulating. Coding using both programs will be considered.**
- **Investigate the effect of the problem parameters on the behavior of the solution.**
- **Write a draft for the project by week 12. (10 points)**
- **Submit the final report and give a presentation by week 14 (70 points)**
- **It is a must to use any available simulation tool (COMOSOL, CST, ANSYS, etc.…) to verify your results.**
- **Comparison with other research outcomes is a plus.**